Removal Note: the strict weight-modulus route to the MPS Fundamental Theorem

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This note records, before its deletion, the strict weight-modulus ("dominant-block peeling") route that the formal development used for the non-periodic Matrix Product State Fundamental Theorem until it was superseded by the sector-coefficient comparison route of [CPGSV16, §II].[1] The purpose of the note is to keep a coherent mathematical record of what the route assumed, what it was designed to prove, why the assumption is not faithful to the source canonical form, and why nothing load-bearing is lost when the route is removed. It complements `docs/paper-gaps/cpsv16_nondominant_per_block_projection.tex`, which records the analytic failure of the projection step in detail.

1 The strict canonical form and the predicates being removed

Fix a finite family of normal MPS site tensors $(A_k)_{k=0}^{r-1}$ with nonzero complex weights μ_k, assembled into the block-diagonal tensor $A = \bigoplus_k \mu_k A_k$. Its matrix-product vector at chain length N expands as

$$V^{(N)}(A) = \sum_{k=0}^{r-1} \mu_k^N \, V^{(N)}(A_k). \tag{1}$$

The removed route strengthened the canonical form by a *strict* ordering of the weight moduli together with a dominant normalization,

$$|\mu_0| > |\mu_1| > \cdots > |\mu_{r-1}|, \qquad |\mu_0| = 1. \tag{2}$$

In the formalization (2) was carried by the predicate HasStrictOrderedNonzeroWeights (field mu˙strict˙anti) and by the canonical-form structure IsNormal-

[1] The replacement route is the one anchored in `blueprint/src/chapter/ch10_bnt.tex` and `ch11_fundamental_theorem_proof.tex`; its Lean home is `TNLean/MPS/FundamentalTheorem/SectorBNT/Fundamental`.

CanonicalFormBNT (fields mu˙strict˙anti and mu˙dom˙norm˙one).[2] The block-separation and "fundamental theorem for the canonical form" declarations built on (2) lived in the now-deleted module `TNLean/PiAlgebra/CanonicalFormSep.lean`; the consuming injectivity-after-blocking results lived in `TNLean/MPS/CanonicalForm/SectorComparison/NormalityChain`.

2 The argument the route was designed to support

The strict ordering (2) was not a cosmetic convention. It was the engine of a dominant-block peeling proof of the equal-MPV uniqueness statement. Given two assembled families A, B with $V^{(N)}(A) = V^{(N)}(B)$ for all N, normalize (1) by the dominant weight μ_0^N. Since $|\mu_k/\mu_0| < 1$ for every $k \neq 0$ under (2), the renormalized contribution of every sub-dominant block tends to zero,

$$\left(\frac{\mu_k}{\mu_0}\right)^N \xrightarrow[N\to\infty]{} 0 \qquad (k \neq 0), \tag{3}$$

so in the limit only the leading block survives. One then projects onto the leading matrix-product vector, matches the leading blocks of A and B, subtracts ("peels") the matched leading block, and recurses on the strictly shorter remaining family. The strict gap in (2) supplies both the unique dominant block at each step and the spectral separation that makes (3) and the induction on the number of blocks terminate. For the equal-MPV case this is sound, because the leading-pair comparison produces the *exact* identity $\mu_0^A = \mu_0^B \zeta$ with $|\zeta| = 1$, after which the leading summand cancels exactly and the tail again satisfies an equality of the same shape.

3 Why strict ordering is not faithful to the source canonical form

The source canonical form of [CPGSV21, Def. 4.2, lines 1864–1884] groups the normal blocks into sectors. A sector indexed by j may contain several copies of one normal tensor with weights $\mu_{j,q}$, so the coefficient attached to a basis-of-normal-tensors vector is the *power sum*

$$c_j(N) = \sum_q \mu_{j,q}^N, \tag{4}$$

not a single scalar power. The strict ordering (2) silently collapses (4) to one term μ_j^N, i.e. it assumes one copy per sector, and it forbids equal-modulus blocks. Both restrictions exclude canonical forms that the source theorem must cover:

[2] HasStrictOrderedNonzeroWeights and IsNormalCanonicalForm.ofStrictSeparatedData in `TNLean/PiAlgebra/CanonicalFormSepAux`; MPSTensor.IsNormalCanonicalFormBNT in `TNLean/MPS/BNT/Construction`.

- $C \oplus (-C)$, with weights 1 and -1 of equal modulus, has coefficient $c(N) = 1 + (-1)^N$, which is not a single scalar power;
- $C \oplus D$ with two distinct non-gauge-equivalent normal tensors and equal weights $(1, 1)$ has two basis-of-normal-tensors vectors of equal modulus;
- $C \oplus \frac{1}{2}C$ is one sector with two copies of unequal modulus, coefficient $1 + (\frac{1}{2})^N$.

Grouping into a basis of normal tensors is governed by gauge-phase equivalence of the block tensors, not by distinctness of weight moduli; equal-modulus sectors are part of the theorem, not an afterthought.

4 Why the peeling argument fails beyond the equal-MPV case

For the proportional statement, where $V^{(N)}(A) = c_N\, V^{(N)}(B)$ with a nonzero scalar sequence c_N, the peeling argument breaks even under (2). Projecting onto a *non-dominant* block B_{k_0} with $k_0 \neq 0$ sends both sides to zero: the left side is a bounded coefficient times a decaying overlap, and the diagonal right-hand term carries the factor $|\mu_{k_0}/\mu_0|^N \to 0$. No contradiction is available, and the exact leading-coefficient identity that the recursion needs is false at finite chain length. This obstruction, and the two independent probes confirming it, are recorded in `docs/paper-gaps/cpsv16_nondominant_per_block_projection.tex`.[3]

5 The replacement and the verdict

The current route does not take limits. At a fixed sufficiently large N it isolates B_{k_0}, observes that the remaining terms lie in the span of $\{V^{(N)}(A_j)\}_j \cup \{V^{(N)}(B_k) : k \neq k_0\}$, and uses eventual linear independence of the full combined family to force the isolated coefficient $c_N \mu_{k_0}^N$ to vanish, contradicting $c_N \neq 0$ and $\mu_{k_0} \neq 0$. This identifies coefficients even when they decay, and it needs no ordering of the moduli. It is realized over the raw sector weights and power-sum coefficients (4) of SectorDecomposition, with the combined-family statement MPSTensor.IsBNTCanonicalForm.combined_family_eventually_li and the modulus-free nonvanishing of the sector coefficient MPSTensor.IsBNTCanonicalForm.coeff_not_eventually_zero.

The verdict is that the strict route was genuinely load-bearing for the old peeling architecture, but that architecture was itself the one-copy specialization just described. The headline Fundamental-Theorem declarations MPSTensor.ft_sector_bnt_equal_global_gauge and MPSTensor.ft_sector_bnt_equal_mps_gaugeEquiv_literal do not depend, by proof term, on (2) or on any declaration of the deleted `TNLean/PiAlgebra/CanonicalFormSep.lean`. Removing the strict route

[3] See also `audits/2026-05-13_cpsv16_ft_discharge_attempt.md` and `audits/2026-05-13_cpsv16_ft_exact_leading_coeff.md`.

therefore deletes a parallel, strictly weaker development; it does not weaken the formalized theorem.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.

[CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.